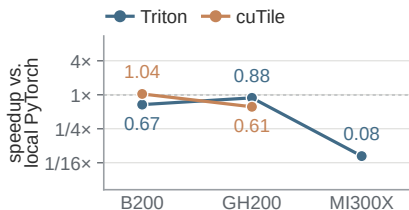
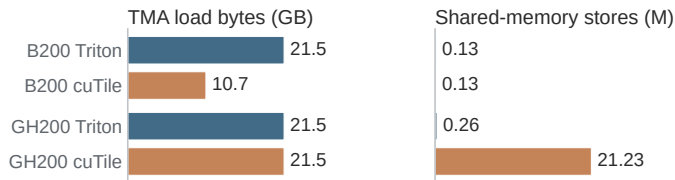


A Matrix operand delivery (matmul, FP32)



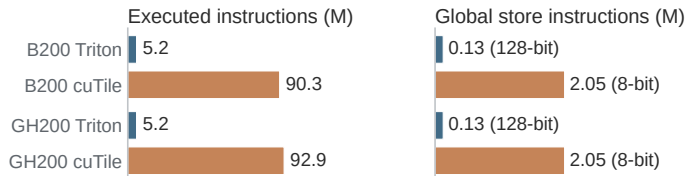
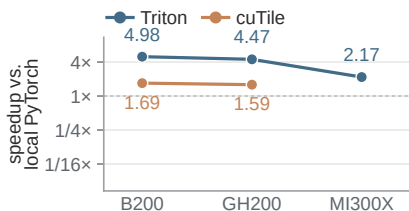
NVIDIA Profiling (NCU)



MI300X (ROCm / ISA evidence): TensorDescriptor loads lowered to 96 scalar 32-bit global_load_dword; MFMA util. 4.5%

Confounders: Triton reads a B operand transposed before timing; cuTile tiles differ (256×256 on B200, 128×128 on GH200)

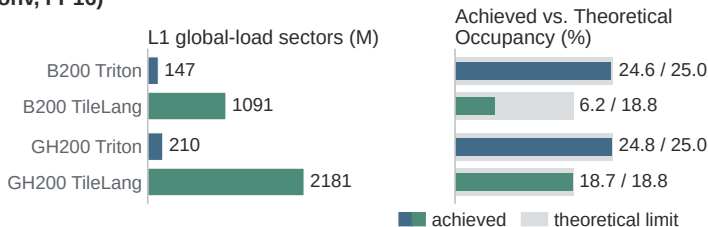
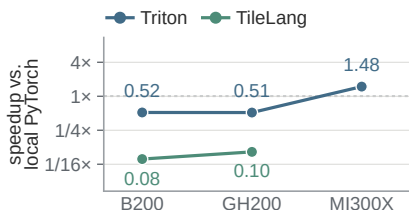
B Indexing overhead (destindex, INT8)



MI300X (ISA evidence): Triton stores each row with 4 per-lane buffer_store_byte, no vector store

Triton and cuTile write the same 2.05 M store sectors on both devices; the PyTorch baseline (index_copy) differs by vendor

C Memory access and latency hiding (1d_conv, FP16)



MI300X (ISA / PyTorch trace): PyTorch runs MIOpen implicit GEMM plus 3 transpose kernels; Triton loads with buffer_load_ushort

NVIDIA Triton and TileLang (B200, GH200): 16-bit loads and equal DRAM reads (0.67 GB); TileLang uses a different kernel body on Hopper